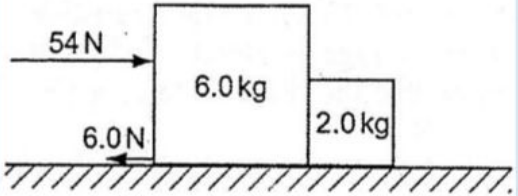
The mass cancels out completely, just like free fall.

6 A force of 54 N pushes two touching blocks of mass 6.0 kg and 2.0 kg along a flat surface. The frictional force between the blocks and the surface is 6.0 N.



What is the force exerted on the 2.0 kg block?

A 12 N **B** 36 N **C** 45 N **D** 48 N

Ans: A

Applying Newton's 2nd law to the entire system: $F_{net} = 54 - 6.0 = 48$ N

$$a = \frac{F_{net}}{m_{total}} = \frac{48}{8.0} = 6.0 \text{ m s}^{-2}$$

Now consider the 2.0 kg block alone. The only horizontal force acting on it is the contact force from the 6.0 kg block pushing it forward (there is no separate friction stated for the 2.0 kg block individually — the 6.0 N friction acts on the whole system).

Applying Newton's 2nd law to the 2.0 kg block:

$$F = ma = 2.0 \times 6.0 = 12 \text{ N}$$